ROTATIONAL DYNAMICS

PPT-3

By- Mrs. P. Joshi (S. B. Patil College Ravet)

Moment of Inertia: (resistance of body to undergo rotational motion)

It is defined as a body's tendency to resist angular acceleration (thus rotational motion)

There are 3 basic types of M.I.

1. Mass Moment of Inertia → Measure of distribution of mass of object relative to given axis
2. Area Moment of Inertia* → Reflects how body's points are distributed relative to given axis
3. Polar Moment of Inertia* → Shaft or Beam's resistance to being distorted by torsion, as a function of its shape.

**Extra information not in syllabus*

Mass Moment of Inertia:

- It is a measure of distribution of the mass of an object relative to a given axis.
- Denoted by I,
- for single particle $I_O = M R^2$, where O is the axis of rotation & R is the distance from axis
- Unit = kg-m^2 Dimension: $[L^2 M^1]$

- Linear motion--- applied force depends on mass
- Rotational motion --- applied force depends on mass and distance from axis.
- We are studying Mass Moment of Inertia which is given by $I_O = M R^2$, where O is the axis of rotation & R is the distance from axis.

Few Properties of M.I.

- M.I. considers the distance wise distribution of mass around the axis of rotation.
- Like other analogies, M.I. in rotational motion is analogous to mass in linear motion.
- Both of these, express a body's tendency to resist motion (angular or linear)

M.I. depends upon following factors: (2mark question with diagram and formula)

i. Mass, shape and size of body.

ii. Distribution of mass in body about an axis of rotation

iii. Position and orientation of axis of rotation.

PHYSICAL SIGNIFICANCE OF M.I.

- To produce linear motion an unbalanced force is applied. Mass of body opposes this motion thus "mass" provides inertia. Given by, $F = ma$
- To produce rotational motion in a body an unbalanced torque is applied. In this case inertia of the body is called rotational inertia or moment of inertia. (given by I)
- The relation between moment of Inertia (I), torque (τ) and angular acceleration (α) is given by, $\tau = I\,\alpha$
- We conclude that, moment of inertia plays same role in rotational motion as the mass of the body does in linear motion.

Analogies between Linear and Rotational Motion:

Linear	Rotational
Linear displacement (x or s)	Angular displacement (θ)
Linear velocity (v)	Angular velocity (ω)
Linear acceleration (a)	Angular acceleration (α)
Force (F)	Torque (τ)
Mass (m)	Moment of Inertia (I)
Translational K.E. = $Mv^2 / 2$	Rotational K.E. = $I\omega^2$

To find the analogy for K.E. in translational (linear) and Rotational motion we consider

Derivation for Rotational Kinetic Energy:

- Rotational K.E. = sum of translational K.E. of all particles present in body.

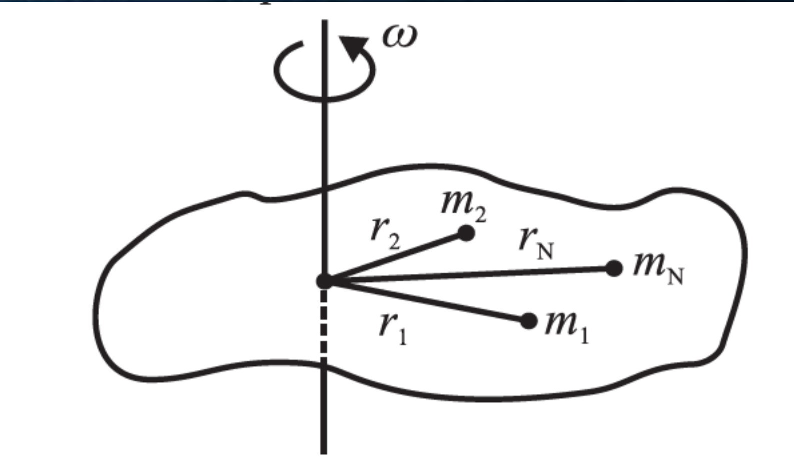


Fig. 1.12: A body of N particles.

Derivation for Rotational Kinetic Energy

Consider a body of random shape and having N particles, of masses m_1, m_2 ….. m_N and distance r_1, r_2 ….. r_N from axis of rotation. Let ω be its angular velocity.

It is a rigid body (distance between any two points inside body is constant) and axis of rotation is perpendicular to paper.

- When the body rotates around its axis, each of these particles perform UCM around this axis with same angular speed ω.
- But, as r (distance from axis) is different for each particle, they will have different translational velocity ($v = r\,\omega$). Let these velocities be v_1, v_2 ….. $v_N = r_N\,\omega$.
- Let us consider translational K.E. for first particle,

$K.E._1 = (m_1 v_1^2)/2$ but $v_1 = r_1\,\omega$ therefore, $K.E._1 = (m_1 r_1^2 \omega^2)/2$

Similarly for all other particles we shall have translational K.E. to be

$$K.E._2 = (m_2 r_2^2 \omega^2)/2\ ,$$
$$K.E._3 = (m_3 r_3^2 \omega^2)/2\ ,$$
So on ………
$$K.E._N = (m_N r_N^2 \omega^2)/2$$

We know,

Rotational K.E. = sum of translational K.E. of all particles present in body.

Therefore,

Rotational K.E. = $(m_1 r_1^2 \omega^2)/2 + (m_2 r_2^2 \omega^2)/2 + \ldots\ldots + (m_N r_N^2 \omega^2)/2$

$= (1/2)\,(m_1 r_1^2 + m_2 r_2^2 + \ldots\ldots + m_N r_N^2)(\omega^2)$

But $m \times r^2 = I$ (moment of Inertia)

Rotational K.E. = $(1/2)\,(I_1 + I_2 + \ldots\ldots + I_N)\,(\omega^2)$

Let $I_1 + I_2 + \ldots\ldots + I_N = m_1 r_1^2 + m_2 r_2^2 + \ldots\ldots + m_N r_N^2 = I$

Rotational K.E. = $(1/2)\,(I)\,(\omega^2) = I\,\omega^2/2$

$$\text{K.E.} = \frac{1}{2} m v^2$$

translation K.E

Thus, in going from translation K.E to Rotational K.E.

- m is replaced by I and
- v by ω

$$\text{K.E.} = \frac{1}{2} I \omega^2$$

rotational K.E

Therefore, *I can be defined as rotational inertia or moment of inertia of the object about the given axis of rotation*.

This $I = \sum m_i r_i^{\,2}$ of an object depends upon,

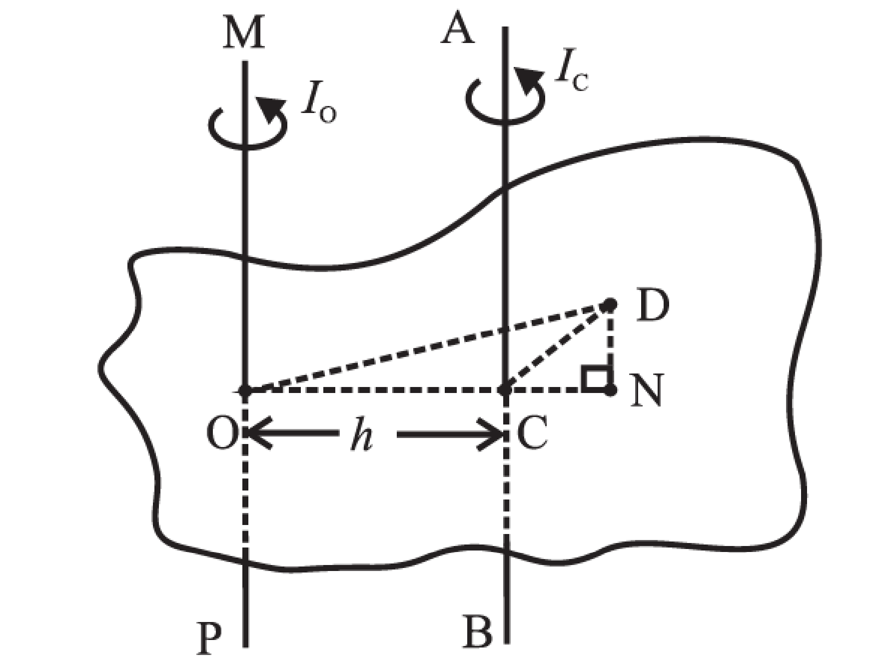
- Individual masses ($m_1, m_2 \ldots\ldots m_N$) and
- Distribution of these masses about the given axis of rotation ($r_1, r_2 \ldots\ldots r_N$)

If we have a different axis, this I will again depend upon on mass distribution around that axis and if the object is not symmetric it will be different for each axis.

i.e. if you have a non-symmetric object like this, you will calculate I separately for axis through point O and axis through point C, as both axes shall have different distribution of masses

- Though by far we considered the object to consist of finite number of particles
- In reality, We usually have a homogenous rigid object.
- So instead of summation, we use integration.

$$I = \int r^2 \, dm$$



Further we will study, M.I. for different objects, like,

1. M.I. of Ring
2. M.I. of Disc

Study in detail

3. M.I. of Sphere
4. M.I. of Rod
5. M.I. of Cylinder
6. M.I. of Right Circular Cone
7. M.I. of Parallelepiped

Formulas Given Directly for each with its cases

$I = \int r^2 \, dm$

If we do not know the mass distribution, theoretically it is not possible to solve the above integral. But we can still find M.I. experimentally.

Let us consider the application of M.I.

1) Moment of Inertia of a uniform ring:

- The mass of this ring is uniformly distributed on its circumference.
- It is a two-dimensional object with negligible thickness.
- If this ring is rotating about its own axis, its entire mass M is practically at a distance equal to its radius R from axis.
- Therefore, expression for its M.I. is

I = (mass) x (sq. of distance from axis)

$I = M R^2$ → For ring

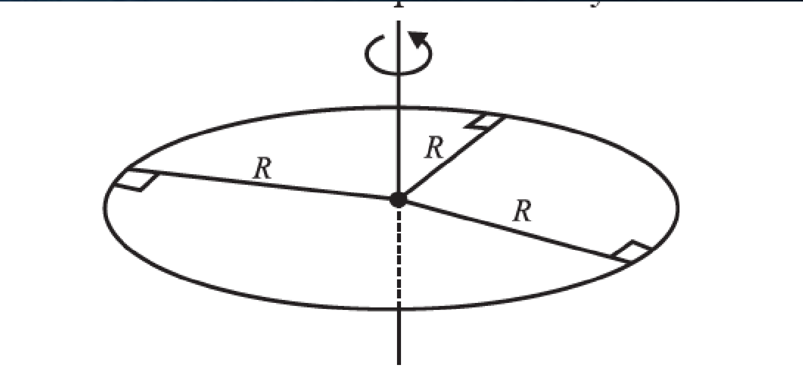
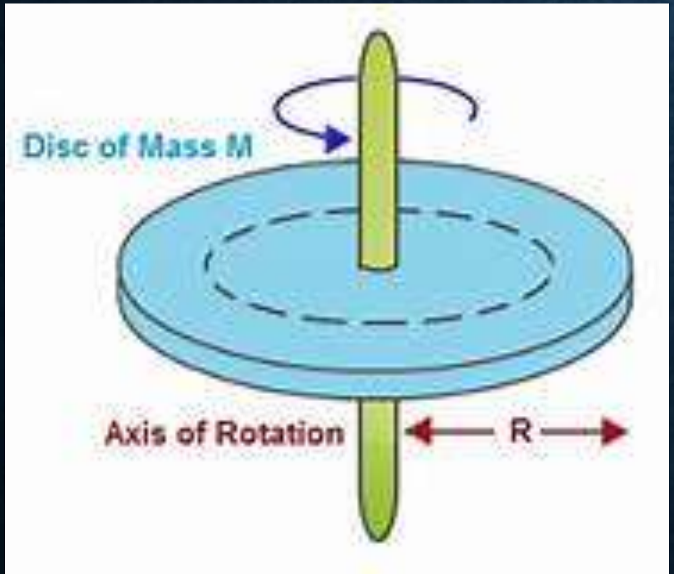


Fig. 1.13: Moment of Inertia of a ring.

2) Moment of Inertia of a Uniform Disc:

Disc Description:

- Two dimensional circular object
- Negligible thickness
- Uniform disc – Mass per unit area (σ) and composition same throughout.
- Disc is rotating about its own axis. This axis is passing through the center of disc.



Surface Density (σ):

- Mass per unit area.
- The ratio σ = mass/ area = m/A is called Surface density
- For uniform disc of mass M and radius R,

  $\sigma = M/\pi R^2$

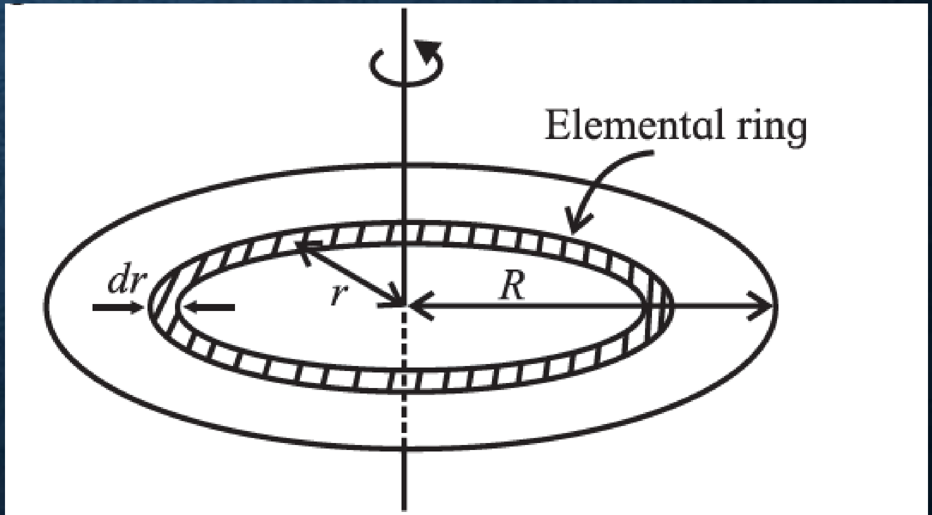


Fig .1.14: Moment of Inertia of a disk.

- Consider the uniform disc given in figure, it has mass M and radius R.
- As it is uniform, we can consider it to be consisted of number of concentric rings whose radii is increasing from zero to R.
- One of such concentric **ring** may have mass *dm* (shaded portion in figure) and width *dr,* very small compared to entire radius of ring
- Practically, this *dr* is considered to be less than least count of the instrument that measures it.

Therefore, Area of ring = $2\pi r.dr$ i.e. $\sigma = dm / 2\pi r.dr$

i.e. $dm = \sigma\, 2\pi r \,.\, dr$

- As this discussion is about the concentric **ring**, this entire mass dm is at a distance r from the axis of rotation. Thus, its M.I. will be

$$I_r = dm\,(r^2)$$

- So, the M.I. of disc will be integration of I_r from 0 to R.

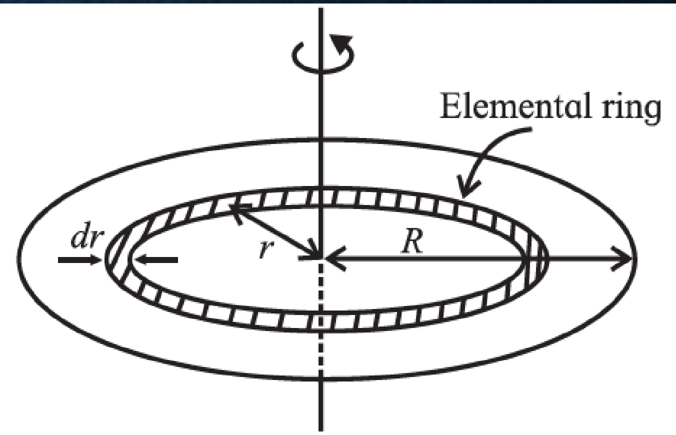


Fig .1.14: Moment of Inertia of a disk.

$$\therefore I = \int_0^R I_r = \int_0^R dm \cdot r^2 = \int_0^R 2\pi\sigma r \cdot dr \cdot r^2 = 2\pi\sigma \int_0^R r^3 \cdot dr$$

$$\therefore I = 2\pi\sigma\left(\frac{R^4}{4}\right) = 2\pi\left(\frac{M}{\pi R^2}\right)\left(\frac{R^4}{4}\right) = \frac{1}{2} MR^2$$

Radius of Gyration:

- As we said before, theoretically you can find M.I. only if you know the mass distribution in object. But experimentally, we can find M.I. for all objects.
- Also, this M.I. depends upon mass of the object and how it is distributed around the axis of rotation.
- So if we only want to know the mass distribution around the axis,

  Let us write M.I. as $I = MK^2$, where M is mass of object and this mass is effectively at a distance K, from the axis of rotation.
- This K, is called as Radius of Gyration.
- It means that the mass of object is effectively at a distance K from the axis of rotation. If K is radius of gyration, $I = MK^2$, is the M.I.
- *Radius of Gyration, can be defined as the distance from the axis of rotation to a point where the total mass of the body is supposed to be concentrated. So if you find M.I. at this point it will be equal to the M.I. that you would get in regular way. (Simple trick !!)*

- Unit of K = metre in S.I. and centimetre in CGS and Dimension = $[L^1]$
- K depends upon shape and size of body.
- *(This K should not be confused with Centre of mass; because 1)K changes with axis 2) K is present only when body is rotating while Centre of mass exists for both stationary and rotating body 3) I about Centre of mass is zero)*

Physical Significance of K:

- For a given object mass (M) always remains constant, what changes is K as per the choice of axis of rotation.
- Thus, K depends upon shape and size of body. It measures the distribution of mass about the axis of rotation.
- For small value of K, we say that mass is distributed close to the axis of rotation, so M.I. is small.
- For larger value of K, mass distribution is at large distance from axis of rotation, so M.I. is large.

Explanation for
how K changes with shape

Consider a uniform ring and a uniform disc, both of the same mass M and same radius R. Let I_r and I_d be their respective moment of inertias.

If K_r and K_d are their respective radii of gyration, we can write,

$$I_r = MR^2 = MK_r^2 \quad \therefore K_r = R \text{ and}$$

$$I_d = \frac{1}{2} MR^2 = MK_d^2 \quad \therefore K_d = \frac{R}{\sqrt{2}} \quad \therefore K_d < K_r$$

It shows mathematically that K is decided by the distribution of mass. In a ring the entire mass is distributed at the distance R, while for a disc, its mass is distributed between 0 and R. Among any objects of same mass and radius, ring has the largest radius of gyration and hence maximum M.I.

- While solving for M.I. , we generally take axis of rotation at axis of symmetry . → *A line that divides any object in half*
- But in some cases this axis may not be at axis of symmetry.
- It can be parallel or perpendicular to the symmetric axis.

For such cases, we need theorem of parallel axis and theorem of perpendicular axis.

1) Theorem of Parallel axes:

In order to apply this theorem to any object, we need two axes parallel to each other with one of them passing through center of mass of object.

Description: Figure shows object of mass(M),

- MOP is random axis passing through O and ACB is axis passing through Centre of mass C.
- MOP is at a distance of h from ACB and they are parallel to each other.

Let us consider small point mass dm at point D.

Construction: We draw perpendicular DN on extended line of OC upto N. Join D point with C and O.

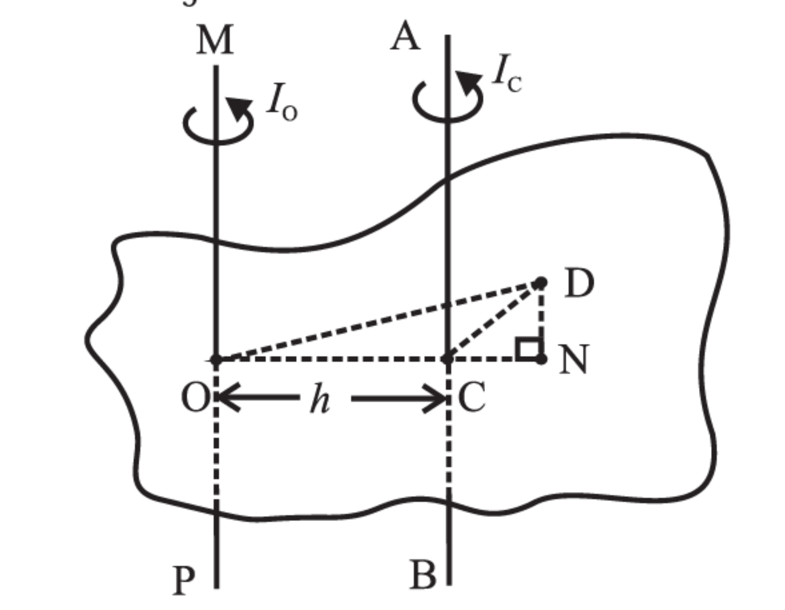


Fig. 1.15: Theorem of parallel axes.

- Considerations: M.I. at point O w.r.to D is $I_O = \int (DO)^2\, dm$ and at point C w.r.to D is $I_C = \int (DC)^2\, dm$

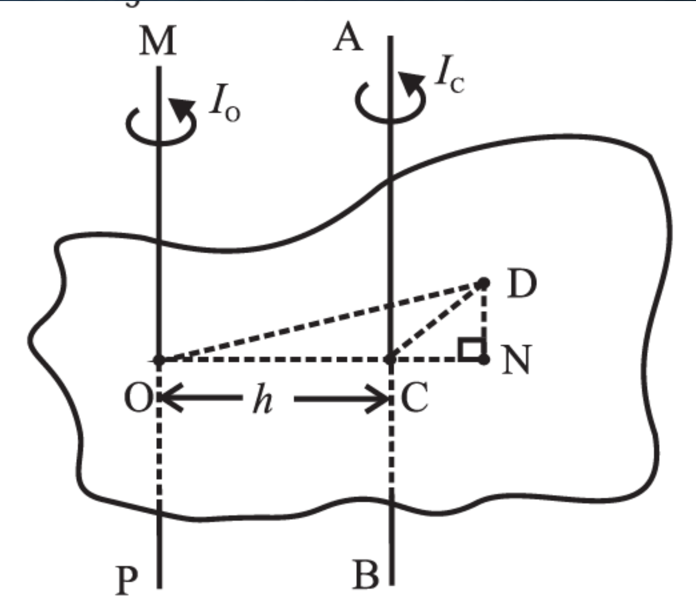


Fig. 1.15: Theorem of parallel axes.

Let us consider I_O (our axis of interest),

$I_O = \int (DO)^2\, dm = \int ([DN]^2 + [NO]^2)dm$ ….. By Pythagoras Theorem

But, $[NO]^2 = [NC + CO]^2 = [NC]^2 + [CO]^2 + [2.NC.CO]$

Therefore, $I_O = \int ([DN]^2 + [NC]^2 + [CO]^2 + [2.NC.CO])\, dm$

But $[DN]^2 + [NC]^2 = [DC]^2$

Therefore, $I_O = \int ([DC]^2 + [CO]^2 + [2.NC.CO])\, dm$

Separating integration to each term and substituting, $CO = h$, and values of I_C and $\int dm = M$,

$I_O = \int [DC]^2\, dm + h^2 \int dm + 2h \int NC\, dm$

$I_O = I_C + M h^2 + 2h \int NC\, dm$

So we have, $I_O = I_C + M h^2 + 2h \int NC\, dm$

This NC is the distance of point from C.M., any mass distribution is symmetric about C.M., thus as per its definition $\int NC\, dm=0$

So we get, $\boxed{I_O = I_C + M h^2}$ $\longrightarrow$ this is mathematical form of theorem of parallel axis.

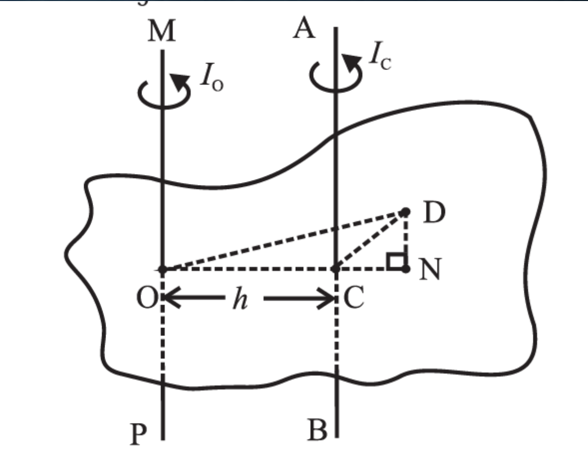


Fig. 1.15: Theorem of parallel axes.

Definition: The theorem of parallel axis states that, the moment of inertia (I_O) of an object about any given axis, is the sum of

- moment of inertia (I_C) about an axis parallel to the given axis, and passing through the Centre of mass *and*
- the product of the mass (M) of the object and the square of the distance between the two axes (h^2)

[if DN falls on OC, i.e. point N coincides with C. In that case , DC will be the distance of point from C.M. , $\int DC\, dm=0$ i.e. therefore, $I_C =0$ hence, $I_O = M h^2$. *So if the point D is taken such that DN falls on OC, it is as good as calculating M.I. for only one axis, which is axis of rotation]*

2) Theorem of Perpendicular Axis:

- This theorem relates M.I. of laminar object about three mutually perpendicular and concurrent axes, two of them in the plane of object and third perpendicular to the object.
- Laminar objects: any two dimensional objects like leaf, ring, disc, etc.
- By far, we considered that the object can rotate only along one direction i.e. z axis, but in this theorem we are considering the possibility that the object can rotate about any axes from x, y and z

(Consider a rupee coin as a disc and try rotating it about x, y and z axes)

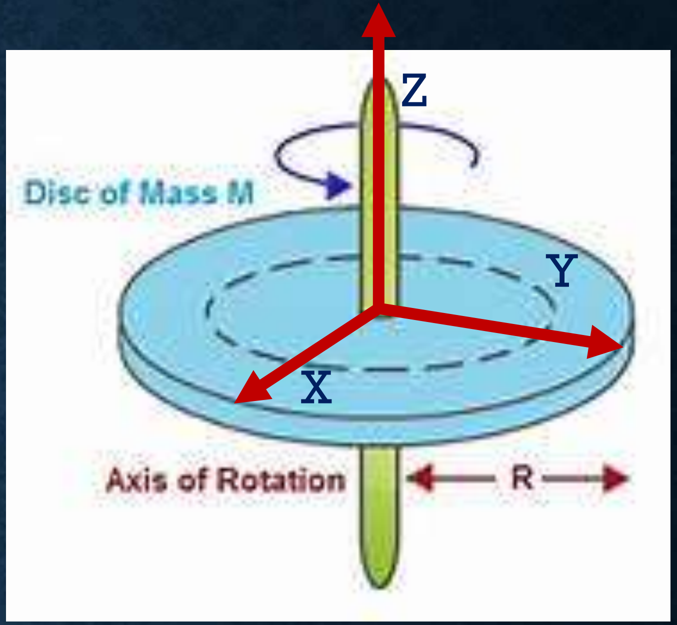


Image Description:
Consider a mass element dm located at point P.
PN = x and PM = y are perpendiculars drawn from P on X and Y axes respectively.
The distance of point P from x-axis is 'y', from y-axis is 'x' and from z-axis is $\sqrt{(x^2 + y^2)}$

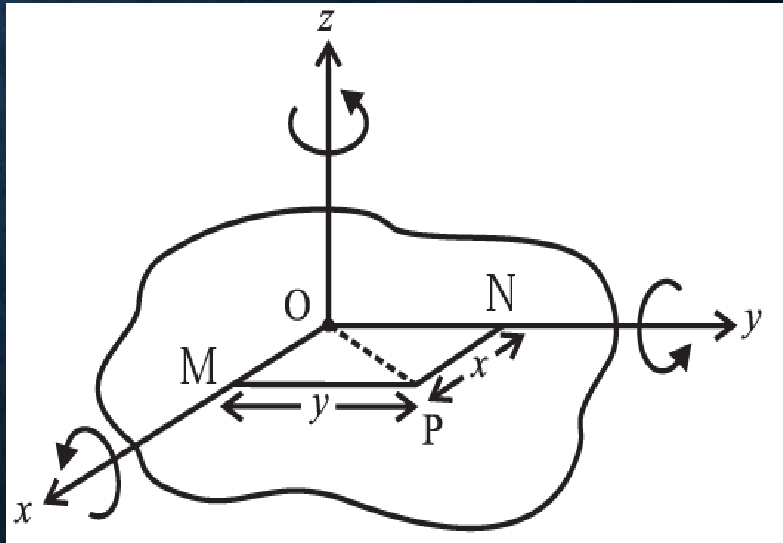


Fig. 1.16: Theorem of perpendicular axes.

We know, I = (mass) x (sq. of distance from axis)

therefore,

$I_x = \int (y^2)\, dm$ $\qquad$ $I_y = \int (x^2) dm$ $\qquad$ $I_z = \int (x^2 + y^2) dm$

I_x , I_y ,and I_z are respective moment of inertias of body about x, y and z axes.

If we consider the expression for I_z , we get

$$I_z = \int (x^2 + y^2) dm$$

$$= \int (x^2)\, dm + \int (y^2) dm$$

$$I_z = I_y + I_x$$

this is mathematical form of theorem of perpendicular axis.

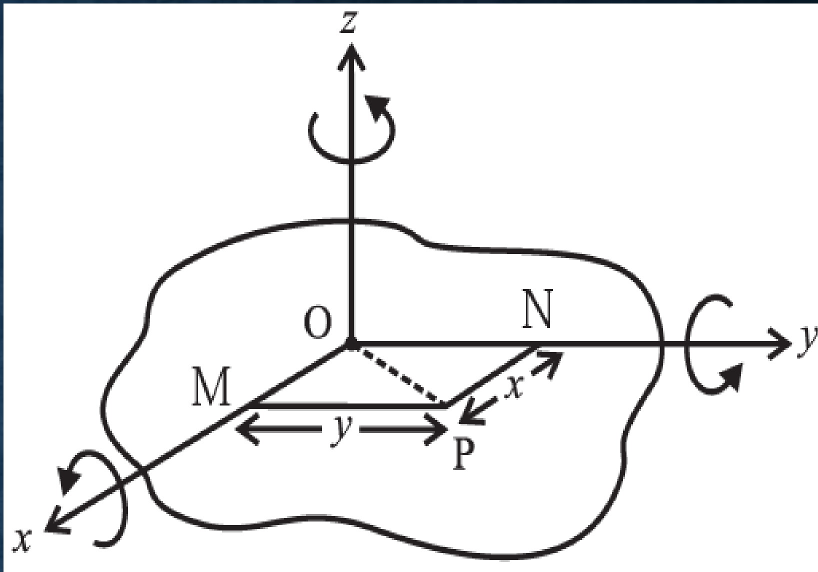


Fig. 1.16: Theorem of perpendicular axes.

Definition: It states that, the M.I. (I_z) of laminar object about an axis (z) perpendicular to its plane is sum of moment of inertias about two mutually perpendicular axes (x and y) in its plane, all the three axes being concurrent.

Angular Momentum

- Also known as moment of linear momentum (*i.e. multiply linear momentum (p=mv) with distance*)
- Analogous to linear momentum.

Definition: If $\bar{p}$ is the instantaneous linear momentum of a particle undertaking a circular motion, its angular momentum at that instance is given by,
$\bar{L} = \bar{r} \times \bar{p}$, where $\bar{r}$ is the position vector from the axis of rotation.

In magnitude,
It is the product of linear momentum and its perpendicular distance from the axis of rotation.
i.e $L = (p)(r \sin\theta)$, where θ is the small angle between the directions of $\bar{p}$ and $\bar{r}$

Unit = $(kg\text{-}m^2)/s$ in S.I. And $(g\text{-}cm^2)/s$ in CGS.
Dimension= $[L^2 M^1 T^{-1}]$

Expression for *Angular Momentum* in terms of M.I.

Description: Consider a body of random shape and having N particles, of masses $m_1, m_2 \ldots\ldots m_N$ and distance $r_1, r_2 \ldots\ldots r_N$ from axis of rotation. Let ω be its constant angular velocity.

- When the body rotates around its axis, each of these particles perform UCM around this axis with same angular speed ω.
- But, as r (distance from axis) is different for each particle, they will have different translational velocity ($v = r\,\omega$). Let these velocities be $v_1, v_2 \ldots\ldots v_N = r_N\,\omega$.

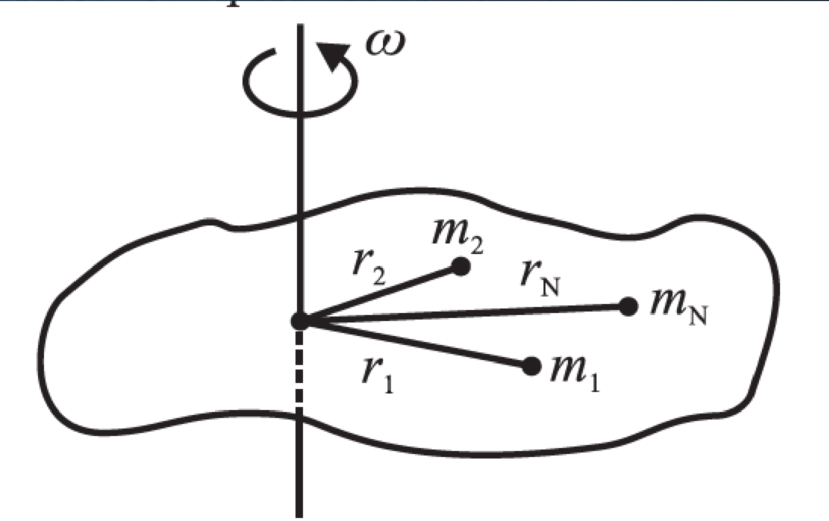


Fig. 1.12: A body of N particles.

- The directions for velocities $v_1, v_2 \ldots\ldots v_N$ are along the tangents to their respective tracks.
- Therefore, the linear momentum for first particle will be

$p_1 = m_1 v_1$ and for further particles, $p_2 = m_2 v_2$, $p_3 = m_3 v_3$ and so on.

Direction of p_1 is along that of v_1 and so on for others.

Therefore, the angular momentum is given as $L_1 = p_1 r_1 = m_1 r_1^{\,2}\,\omega$

$L_2 = p_2 r_2 = m_2 r_2^{\,2}\,\omega$

$L_3 = p_3 r_3 = m_3 r_3^{\,2}\,\omega$

So on, $L_N = p_N r_N = m_N r_N^{\,2}\,\omega$

- For a rigid body with fixed axis of rotation,
  - All these angular momentum are directed along the axis of rotation, and
  - Its direction is given by right hand thumb rule.

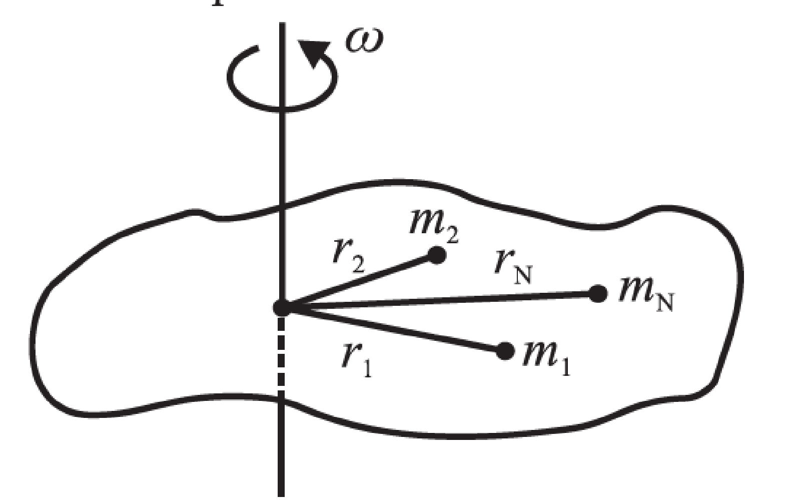


Fig. 1.12: A body of N particles.

As all L have same directions, their magnitudes are directly added.

$L = m_1 r_1^2\omega + m_2 r_2^2\omega + \ldots\ldots + m_N r_N^2\omega$

$= (m_1 r_1^2 + m_2 r_2^2 + \ldots\ldots + m_N r_N^2)\ \omega$

Therefore, as $I = m r^2$

$L = (I_1 + I_2 + \ldots\ldots + I_N)\ \omega$

If we put $I_1 + I_2 + \ldots\ldots + I_N = I$

$L = I\omega$, where I is moment of Inertia about the axis of rotation.

We can see that above equation of

$L = I\omega$ is analogous to $p = m v$

moment of inertia I, replaces mass m as per its physical significance.

The two derivations that we do talk about expressing

Angular momentum and Torque → in terms of Moment of Inertia

Why we do this?

In *linear motion*,

- we first find position of particle (x), then
- the force (f) which caused motion of this particle.
- Once we know the force, we are interested in knowing how this force relates to mass and velocity together, i.e. momentum (p)

↓

The most important property of that particle (or object) which can affect both f as well as p, is mass (M).

In case of *Rotational Motion*,

- We know the angular position of particle on circle from angle (θ),
- To know efforts needed to carry out this circular motion we found out torque, τ (analogous to force)
- Once we know torque τ, we found out its relation with angular momentum, L (analogue of linear momentum)

↓

Similarly, the most important property of object which can affect circular motion, τ and L is Moment of Inertia (I).
Therefore, we are finding τ and L in terms of M.I.

Expression for *Torque* in terms of M.I.

Description: Consider a rigid body of random shape and having N particles, of masses $m_1, m_2 \ldots\ldots m_N$ and distance $r_1, r_2 \ldots\ldots r_N$ from axis of rotation.
This body is rotating with constant angular acceleration (α)

(By far we considered ω and now we are considering α , this is because both ω and α are along same line for anticlockwise rotation of body.
Hence we can talk about ω or α as per requirement.
Another reason for considering α is that in present scenario we are interested in force analogue torque.*)*

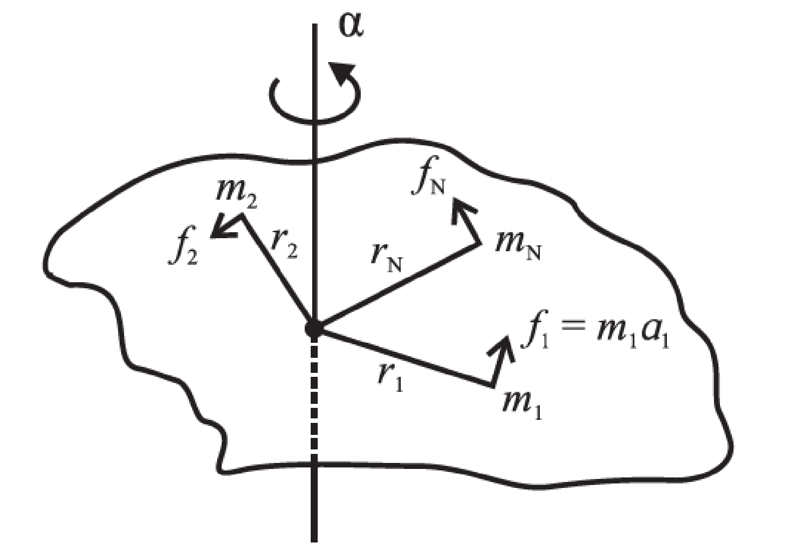


Fig 1.17: Expression for torque.

As the object rotates, each of these particles perform circular motion (not UCM..!) with same angular acceleration but different linear acceleration.
Therefore we have, $a_1 = r_1\,\alpha$, $a_2 = r_2\,\alpha$, so on $a_N = r_N\,\alpha$
Now,
The force experienced by first particle is $f_1 = m_1\,a$, $f_2 = m_2\,a_2$ so on $f_N = m_N\,a_N$

$$= m_1\,r_1\,\alpha \qquad = m_2\,r_2\,\alpha \qquad = m_N\,r_N\,\alpha$$

As these forces are tangential, their respective perpendicular distances from axis of rotation are, $r_1, r_2 ..... r_N$

Thus, the torque experienced by first particle is given as

$\tau_1 = f_1 r_1 = (m_1 r_1 \alpha)(r_1) = m_1 r_1{}^2 \alpha$,

$\tau_2 = m_2 r_2{}^2 \alpha$,

So on

$\tau_N = m_N r_N{}^2 \alpha$

If we restrict motion to a single plane, direction of all these torques are the same, and along same axis

So, their magnitudes can be added,

$\tau = \tau_1 + \tau_2 + ... + \tau_N$

$=(m_1 r_1{}^2 + m_2 r_2{}^2 + ... + m_N r_N{}^2)\alpha$

$= (I_1 + I_2 + ... + I_N)\alpha$

where, I denotes moment of inertia for each particle.

If we put $I_1 + I_2 + ... + I_N = I$

In general, we get

$\tau = I\alpha$

Comparing this to $f = ma$, we know that torque (τ) is analogous to force (f)

Conservation of Angular Momentum:

Just like linear momentum of an isolated system remains conserved in absence of an external unbalanced force. Similarly, we can show that Angular momentum also remains conserved.

We know, $\bar{L} = \bar{r} \times \bar{p}$

where $\bar{r}$ is the position vector from the axis of rotation and $\bar{p}$ is linear momentum.

Differentiating w. r. to time,

Therefore, we conclude that,

- "angular momentum remains conserved in absence of an external torque".
- This is the statement for principle of conservation of angular momentum,
- analogous to conservation of linear momentum.

Differentiating with respect to time, we get,

$$\frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{p}) = \vec{r} \times \frac{d\vec{p}}{dt} + \frac{d\vec{r}}{dt} \times \vec{p}$$

Now, $\frac{d\vec{r}}{dt} = \vec{v}$ and $\frac{d\vec{p}}{dt} = \vec{F}$.

$$\therefore \frac{d\vec{L}}{dt} = \vec{r} \times \vec{F} + m(\vec{v} \times \vec{v})$$

Now $(\vec{v} \times \vec{v}) = 0$

$$\therefore \frac{d\vec{L}}{dt} = \vec{r} \times \vec{F}$$

But $\vec{r} \times \vec{F}$ is the moment of force or torque $\vec{\tau}$.

$$\therefore \vec{\tau} = \frac{d\vec{L}}{dt}$$

Thus, if $\vec{\tau} = 0$, $\frac{d\vec{L}}{dt} = 0$ or $\vec{L} = \text{constant}$.

Examples of Conservation of Angular Momentum:

- In all these applications, the product $L = I\omega$ is constant (once the player acquire certain speed).
- Thus, if moment of inertia I is increased,
  the angular speed ω *decreases* and hence
  the frequency of revolution *n* *decreases.*
- Vice versa, if inertia I is decreased, angular speed ω and frequency n increases.

1) *Ballet Dancer*:

- In this, the dancers have to undertake rounds of smaller and larger radii.

- Dancers come together while taking smaller radius,
  in this case the M.I. of system becomes minimum and
  the frequency increases, to make it thrilling.

- While outer rounds, the dancers outstretch their legs and arms, thus larger radii,
  this increases M.I. thus reducing their angular speed and hence linear speed.
  This helps prevent slipping.

2) Diving in a swimming pool (during competition):

- While on diving board, the divers stretch their body so as to increase the M.I.
- Immediately after leaving the board, they fold their bodies, which reduces M.I. considerably. As a result, the frequency increases and they can complete more rounds in air to make the show attractive.
- Again, while entering the water they stretch their body into a streamline shape, this allows them a smooth entry into the water.

ROLLING MOTION

- Throughout chapter we considered objects moving in circular motion but at one position only, there was no linear motion involved in our discussions.
- In case of Rolling motion, Circular and Linear motion are taking place simultaneously.

- For deriving formula, we theoretically consider that Rolling Motion is result of

1) Circular motion of body as a whole, about its own symmetry axis and
2) Linear motion of body assuming it to be concentrated at its C.M. (i.e. C.M. is performing linear motion)

Accordingly, the object now has two types of K.E., rotational and linear, sum of which makes the total energy. (*During motion P.E. is zero*)

We go ahead to derive the equation for this total energy (E).

Let us consider an object of moment of inertia I, rolling uniformly. Following quantities can then be related.

v = linear speed of C.M. ;

R = Radius of body;

ω = angular speed of the body ($v = R\omega$)

M= mass of body.

K = radius of gyration ; $I = MK^2$

Total kinetic energy of rolling = Translational K.E. + Rotational K.E.

$$\therefore E = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

$$= \frac{1}{2}Mv^2 + \frac{1}{2}\left(MK^2\right)\left(\frac{v}{R}\right)^2$$

$$= \frac{1}{2}Mv^2\left(1 + \frac{K^2}{R^2}\right) \qquad \text{--- (1.18)}$$

- Static friction is essential in this purely rolling motion as it prevents sliding.
- In reality, we always have some slipping involved with rolling motion.

Since the dynamics involved with circular motion present in rolling motion are same as studied before, here we focus on deriving the expression of velocity and acceleration for Rolling Motion.

Figure 1.18 shows a rigid object of mass m and radius R, rolling down an inclined plane, without slipping. Inclination of the plane with the horizontal is θ.

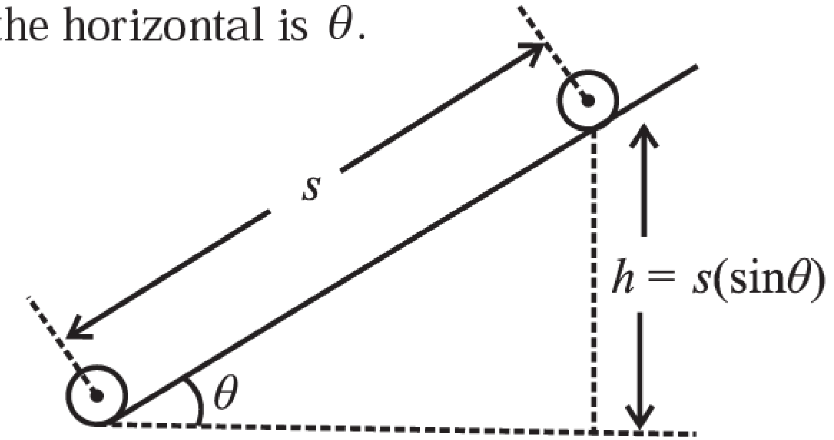


Fig. 1.18: Rolling along an incline.

As motion starts, Gravitational P.E. gets converted into Rolling K.E.

$$\therefore\ E = mgh = \frac{1}{2} M v^2 \left(1 + \frac{K^2}{R^2}\right)$$

m in mgh should be M (printing mistake)

$$\therefore\ v = \sqrt{\frac{2gh}{\left(1 + \frac{K^2}{R^2}\right)}}$$

To find acceleration a, we make use of kinematic equation $2as = v^2 - u^2$

Linear distance travelled along the plane is

$s = h / \sin\theta$

$$2a\left(\frac{h}{\sin\theta}\right) = \frac{2gh}{\left(1 + \frac{K^2}{R^2}\right)} - 0 \quad \Longrightarrow \quad \therefore\ a = \frac{g\sin\theta}{\left(1 + \frac{K^2}{R^2}\right)}$$

For pure sliding, $a = g\sin\theta$ and $v = \sqrt{(2gh)}$, thus the otherwise term in denominator is purely rolling.

Remarks:

1) If expression for M.I. is of the form $n(MR^2)$, then the numerical factor

   n gives value of K^2/R^2.

2) When a rod rolls, it is actually a cylinder that is rolling.

3) While Rolling, the ratio of

Translational K.E. : Rotational K.E. : Rolling K.E. =

$$1 : \frac{K^2}{R^2} : \left(1 + \frac{K^2}{R^2}\right)$$

In Rolling, 60% of energy is translation while 40% is rotational.